UNIVERSITY OF DERBY

College of Science and Engineering

Practice Paper A Solutions

OPERATIONAL RESEARCH – 5CM522

1. Linear Programming

Let D be desks and S be shelving units.

(a) A suitable model is

$$\max Z = 58D + 46S$$

subject to

$$5D + 3S \leq 120$$

$$2D + 4S \leq 96$$

$$S \geq 8$$

$$D, S \geq 0$$

(b) At the optimum, carpentry and finishing are both binding:

$$5D + 3S = 120, \qquad 2D + 4S = 96$$

Hence

$$D = \frac{96}{7} \approx 13.71, \qquad S = \frac{120}{7} \approx 17.14$$

Maximum contribution:

$$Z = 58D + 46S = 1584$$

(c) Binding constraints: carpentry and finishing.

Slack values:

$$5D + 3S = 120 \Rightarrow 0 \text{ carpentry slack}$$

$$2D + 4S = 96 \Rightarrow 0 \text{ finishing slack}$$

$$S - 8 = \frac{64}{7} \approx 9.14$$

So the minimum shelving requirement is not binding.

(d) If shelving contribution rises to £52, the objective becomes

$$\max Z = 58D + 52S$$

The optimal corner remains the same:

$$(D, S) = \left(\frac{96}{7}, \frac{120}{7}\right)$$

with revised contribution

$$Z = \frac{11808}{7} \approx 1686.86$$

So the basis does not change.

(e) The new condition is

$$S \geq 0.3(D + S)$$

which simplifies to

$$3D - 7S \leq 0$$

At the original optimum,

$$\frac{S}{D+S} = \frac{17.14}{30.86} \approx 55.6\%$$

so the original solution remains feasible. The optimal plan is unchanged.

2. Binary Integer Programming

Let $x_A, \ldots, x_F \in \{0, 1\}$ where $x_i = 1$ if project i is selected.

(a) A suitable model is

$$\max B = 14x_A + 17x_B + 11x_C + 20x_D + 9x_E + 15x_F$$

subject to the budget constraint

$$6x_A + 8x_B + 5x_C + 9x_D + 4x_E + 7x_F \leq 21$$

and the logical constraints

$$x_D \leq x_B$$

$$x_A + x_F \leq 1$$

$$x_C + x_E \geq 1$$

$$x_i \in \{0, 1\}$$

(b) The optimal solution is

$$x_B = x_D = x_E = 1, \qquad x_A = x_C = x_F = 0$$

Total cost:

$$8 + 9 + 4 = 21$$

Maximum benefit:

$$17 + 20 + 9 = 46$$

(c) Binary variables are needed because a project cannot be partly selected in any practical sense. A continuous LP could return fractional values such as "0.4 of project D", which would not be meaningful here.

(d) Add

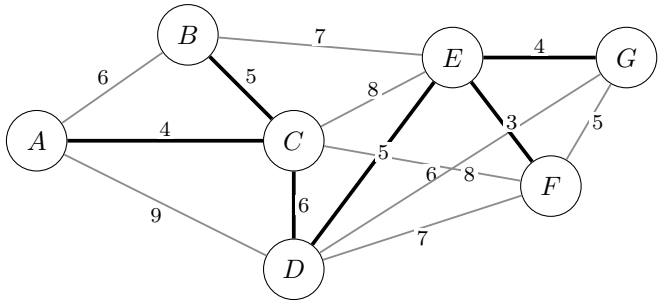
$$x_A + x_B + x_C + x_D + x_E + x_F = 4$$

There is no feasible solution satisfying all rules, the budget and the requirement to choose exactly four projects.

(e) Example extension: allow projects to be staged over several years with different annual budgets and inter-year dependencies.

3. Minimum Spanning Tree

(a) A minimum spanning tree connects all nodes at minimum total cost without forming cycles.



(b) One correct Kruskal solution selects:

$$EF(3),\ AC(4),\ EG(4),\ BC(5),\ DE(5),\ CD(6)$$

(c) Total installation cost:

$$3 + 4 + 4 + 5 + 5 + 6 = 27$$

(d) The cheapest extra non-tree link has cost 5, for example FG.

So one cheapest backup addition is FG, giving revised total cost

$$27 + 5 = 32$$

(e) Limitation: an MST gives no built-in redundancy, so a single link failure can disconnect part of the network.

4. Simulation

(a) A suitable simulation model would treat weekly deliveries and weekly demand as random inputs. State variables would include opening stock, ending stock and overflow stock. Key outputs would be average weekly cost, overflow frequency and the average size of overflow inventory.

(b) One valid 12-week simulation can be produced using seed 2026, giving:

Deliveries:

$$7, 8, 8, 9, 7, 7, 8, 8, 9, 8, 9, 9$$

Demands:

$$8, 7, 6, 7, 5, 8, 7, 8, 7, 7, 6, 6$$

Assume opening stock is 0 and ending stock carries forward week to week.

For capacity 7:

Week	Open	Delivered	Demand	Sales	End Stock	Overflow	Cost
1	0	7	8	7	0	0	126
2	0	8	7	7	1	0	126
3	1	8	6	6	3	0	126
4	3	9	7	7	5	0	126
5	5	7	5	5	7	0	126
6	7	7	8	8	6	0	126
7	6	8	7	7	7	0	126
8	7	8	8	8	7	0	126
9	7	9	7	7	9	2	186
10	9	8	7	7	10	3	216
11	10	9	6	6	13	6	306
12	13	9	6	6	16	9	396

(c) Total 12-week cost for capacity 7:

$$2112$$

Estimated average weekly cost:

$$\frac{2112}{12} = 176.0$$

(d) Using the same 12-week random stream:

Capacity	Total Cost	Average Weekly Cost
6	2106	175.5
7	2112	176.0
8	2208	184.0
9	2304	192.0

On this run, capacity 6 gives the lowest average weekly cost of £175.5. The other averages are £176.0 for capacity 7, £184.0 for capacity 8 and £192.0 for capacity 9. Therefore, 6 bays appears to be the optimal capacity from this simulation. Increasing capacity beyond 6 raises the fixed bay cost faster than it reduces overflow cost on this run.

(e) Two reasons to run a longer or repeated simulation:

The 12-week result may be strongly affected by random variation.

Longer runs or replications would give a more stable estimate of average cost and better evidence for comparing capacities.

5. Forecasting

(a) Using periods 1–12 only, centred 4-quarter moving averages are available for periods 3–10:

Period	Demand	CMA
3	570	602.25
4	690	612.25
5	548	622.25
6	662	633.50
7	608	645.25
8	742	656.25
9	590	666.75
10	708	677.75

A linear trend fitted to these centred moving averages is

$$\hat{T}_t = 568.75 + 10.89t$$

where $t = 1$ corresponds to 2020 Q1.

(b) Seasonal indices:

Additive:

$$Q1 = -75.72, \quad Q2 = 29.16, \quad Q3 = -34.97, \quad Q4 = 81.53$$

Multiplicative:

$$Q1 = 0.8826, \quad Q2 = 1.0446, \quad Q3 = 0.9442, \quad Q4 = 1.1286$$

(c) Validation forecasts for 2023:

Quarter	Actual	Additive Forecast	Multiplicative Forecast
2023 Q1	636	634.60	626.93
2023 Q2	760	750.36	753.38
2023 Q3	692	697.13	691.23
2023 Q4	852	824.52	838.54

Using MAD:

$$\text{MAD}_{\text{add}} \approx 10.91, \qquad \text{MAD}_{\text{mult}} \approx 7.48$$

where absolute error is calculated as $|\text{Actual} - \text{Forecast}|$.

For the multiplicative model:

$$|636-626.93| = 9.07, \quad |760-753.38| = 6.62, \quad |692-691.23| = 0.77, \quad |852-838.54| = 13.46$$

so

$$\text{MAD}_{\text{mult}} = \frac{9.07 + 6.62 + 0.77 + 13.46}{4} \approx 7.48$$

For the additive model:

$$|636-634.60| = 1.40, \quad |760-750.36| = 9.64, \quad |692-697.13| = 5.13, \quad |852-824.52| = 27.48$$

so

$$\text{MAD}_{\text{add}} = \frac{1.40 + 9.64 + 5.13 + 27.48}{4} \approx 10.91$$

The multiplicative model is better.

(d) Forecasts for 2024 using the multiplicative model:

Quarter	Forecast
2024 Q1	665.38
2024 Q2	798.88
2024 Q3	732.35
2024 Q4	887.70

6. Transhipment

(a) Use a network flow model with variables defined on the links of the network.

Let the node sets be:

$$S = \{D_1, D_2, D_3\}, \qquad H = \{H_1, H_2\}, \qquad C = \{C_1, C_2, C_3, C_4\}$$

and let the arc set be

$$A = \{(i,h) : i \in S,\ h \in H\} \cup \{(h,j) : h \in H,\ j \in C\}.$$

Let

$$x_{uv} = \text{flow on arc } (u,v) \text{ for each } (u,v) \in A$$

and let k_{uv} be the corresponding unit transportation cost.

Objective:

$$\min Z = \sum_{(u,v)\in A} k_{uv} x_{uv}$$

Supply constraints for each depot $i \in S$:

$$\sum_{h\in H} x_{ih} \le s_i$$

Demand constraints for each clinic $j \in C$:

$$\sum_{h\in H} x_{hj} = r_j$$

Flow-balance constraints for each hub $h \in H$:

$$\sum_{i\in S} x_{ih} - \sum_{j\in C} x_{hj} = 0$$

Non-negativity:

$$x_{uv} \ge 0 \qquad \text{for all } (u,v) \in A$$

This is the same network-flow style used in Pyomo when variables are indexed over arcs and flow conservation is written at each transhipment node.

(b) One optimal shipment plan is:

Link	Quantity	Unit Cost
$D_1 \to H_2$	85	7
$D_2 \to H_1$	60	4
$D_2 \to H_2$	30	6
$D_3 \to H_2$	110	3
$H_1 \to C_1$	60	4
$H_2 \to C_2$	70	4
$H_2 \to C_3$	80	3
$H_2 \to C_4$	75	5

Total transportation cost:

2480

Unused supply: 15 units at D_1.

(c) Revised model changes:

$$\text{flow through } H_1 \le 130$$

and

$$x_{H_2,C_3} = 0.$$

One revised optimal plan is:

Link	Quantity	Unit Cost
$D_1 \to H_1$	40	5
$D_1 \to H_2$	45	7
$D_2 \to H_1$	90	4
$D_3 \to H_2$	110	3
$H_1 \to C_1$	50	4
$H_1 \to C_3$	80	5
$H_2 \to C_1$	10	6
$H_2 \to C_2$	70	4
$H_2 \to C_4$	75	5

Revised total cost:

$$2520$$

(d) With 20 extra units at H_1, the cost falls back to 2480. One optimal link-flow solution is:

Link	Quantity	Unit Cost
$D_1 \to H_1$	85	5
$D_2 \to H_1$	55	4
$D_2 \to H_2$	35	6
$D_3 \to H_2$	110	3
$H_1 \to C_1$	60	4
$H_1 \to C_3$	80	5
$H_2 \to C_2$	70	4
$H_2 \to C_4$	75	5

Saving:

$$2520 - 2480 = 40$$

Extra capacity cost:

$$50$$

So it is not worthwhile.